

LECTURE #5

Econometrics I

MULTIPLE REGRESSION ANALYSIS INFERENCE

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In the previous lecture #4

- ▶ We derived the OLS estimator for multiple regression models:

$$\hat{\beta}^{OLS} = (X^T X)^{-1} X^T Y.$$

- ▶ 'Partial' interpretation: $\hat{\beta}_j = \frac{\Delta \hat{y}}{\Delta x_j}$, holding all other $x_{\neq j}$ fixed.
- ▶ We listed **four MLR assumptions** $\Rightarrow \mathbb{E}(\hat{\beta}) = \beta$.
- ▶ We discussed model overspecification (**irrelevant** variables) and underspecification (**omitted** variables).
- ▶ **MLR.5 Homoskedasticity:** $\text{Var}(u|x_1, \dots, x_k) = \sigma^2 \mathbb{I}_n \Rightarrow$

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)} \quad \text{or} \quad \text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

- ▶ We finally estimated $\hat{\sigma}^2 = \frac{1}{n-k-1} \sum_{i=1}^n \hat{u}_i^2 \Rightarrow se(\hat{\beta}_j)$.
- ▶ Readings for lecture #5:
 - ▶ Chapter 3: 3.5, Chapter 4: 4.1–4.3, **4.4 (mandatory for/after seminars)**.

Outline

Efficiency of the OLS estimator

Sampling distributions of the OLS estimator

Testing hypotheses about a single population parameter

- t test

- Economic vs. statistical significance

- Confidence intervals

Testing hypotheses about a single linear combination of parameters

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Gauss-Markov theorem

Under **MLR.1** through **MLR.5**, the OLS estimator is the **best linear unbiased estimator (BLUE)**.

- ▶ We know OLS is an 'E'estimator.
- ▶ We know it is 'U'nbiased.
- ▶ 'Linear' estimator means that it can be expressed as a linear function of (the observations of) the dependent variable:

$$\hat{\beta}_j = \sum_{i=1}^n w_{ij} y_i,$$

where w_{ij} can be a function of the sample values of all independent variables.

- ▶ 'Best' here refers to the one with the lowest variance (within this specific family of estimators), i.e., the efficient one.

Linear estimator

- ▶ We show that OLS is a linear estimator for the simple regression case.
- ▶ It means that the estimator can be written as $\hat{\beta}_j = \sum w_{ij}y_i$, i.e., it is a linear combination of y_i , $i = 1, \dots, n$.
- ▶ Only SLR.3 is needed, trivially, for OLS to be linear!
- ▶ To show this, we rewrite the OLS estimator as

$$\hat{\beta}_1 \stackrel{\text{SLR.3}}{=} \frac{\sum y_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = y_1 \frac{x_1 - \bar{x}}{\sum (x_i - \bar{x})^2} + \dots + y_n \frac{x_n - \bar{x}}{\sum (x_i - \bar{x})^2}.$$

- ▶ We thus have weights $w_{11} = \frac{x_1 - \bar{x}}{\sum (x_i - \bar{x})^2}, \dots, w_{n1} = \frac{x_n - \bar{x}}{\sum (x_i - \bar{x})^2}$ and OLS is a linear estimator.
- ▶ Keep in mind that, in general, a specific weight w_{ij} can be zero; it can also be a constant or the same value across more i as long as it does not contain the dependent variable.

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Normality assumption

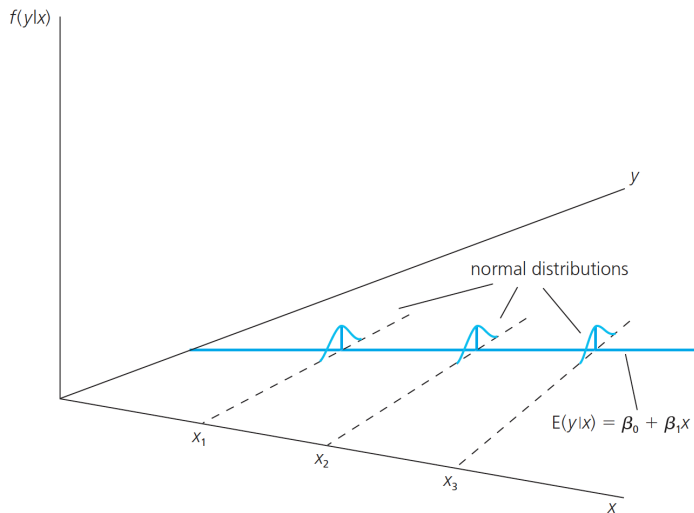
- ▶ We know the expected value and variance of the OLS estimator.
- ▶ However, the shape of the distribution has not been defined or inferred yet.
- ▶ As we condition on $x_1, \dots, x_k$ (i.e., they are not treated as random variables), the distribution of u becomes crucial.
- ▶ **MLR.6 Normality:** The population error u is **independent** of the explanatory variables $x_1, \dots, x_k$ and is **normally** distributed with zero mean and variance σ^2 , i.e., $u \sim N(0, \sigma^2)$.
- ▶ **MLR.6** covers **MLR.4** and **MLR.5**.
- ▶ Assumptions **MLR.1** through **MLR.6** are called the **classical linear model (CLM) assumptions**.

BUE and beyond

- ▶ Under **MLR.1** through **MLR.6**, the OLS estimator is **BUE**, i.e., the **best unbiased estimator**.
- ▶ Conditioning on X and assuming normality of u implies

$$y|X \sim N(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k, \sigma^2).$$

BUE and beyond: Example



Source: Wooldridge (2012)

BUE and beyond

- ▶ **Central limit theorem** $\Rightarrow$ because the error term u is the sum of many different unobserved factors affecting y , we can conclude that it has an **approximate normal distribution**.
- ▶ Assuming normality can be troublesome:
 - ▶ 'CLT argument' for u has some weaknesses:
independence/uncorrelatedness, different/various distributions, additivity, etc.,
 - ▶ truncated distributions/variables,
 - ▶ data (functional) transformations sometimes help:
in economics and finance, the logarithmic transformation is the most popular one.

Normal sampling distribution

- ▶ Normality of u translates into normal sampling distribution of the OLS estimator.
- ▶ Under the **CLM assumptions MLR.1** through **MLR.6**, conditional on the sample of the independent variables,

$$\hat{\beta}_j \sim N(\beta_j, \text{Var}(\hat{\beta}_j)).$$

- ▶ Therefore,

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim N(0, 1).$$

- ▶ However, this holds only if we know σ^2 , which is not the case in most cases.

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t distribution for the standardized estimators

- Under the CLM assumptions MLR.1 through MLR.6,

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1} = t_{df}, \quad (1)$$

where $k + 1$ is the number of unknown parameters in the population model (including the intercept), and $n - k - 1$ is the degrees of freedom (df).

- Note the difference between sd and se .

► William Sealy Gosset

t ratio

- ▶ Most commonly tested null hypothesis

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0.$$

- ▶ **Important:** we are testing the value of the population parameter β_j (without 'hat'), not of its estimate!
- ▶ Under $H_0 : \beta_j = 0$, equation (1) gives us the **t ratio**

$$\boxed{t_{\hat{\beta}_j}} \equiv \frac{\hat{\beta}_j - 0}{se(\hat{\beta}_j)} = \boxed{\frac{\hat{\beta}_j}{se(\hat{\beta}_j)}}.$$

- ▶ **t ratio** identifies the statistically significant independent variables, i.e., the ones whose partial effect (after controlling for all other independent variables) is statistically significantly different from zero.

Hypothesis testing using the t ratio: t test

t **ratio** construction gives us the standard testing framework you **know from Statistics**, i.e., it can be used for testing:

- ▶ One-sided alternatives, e.g.: $H_0 : \beta_j = (\leq) 0$ vs. $H_1 : \beta_j > 0$.
- ▶ Two-sided alternative: $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$.
- ▶ Through the **critical value(s)**:
significance level $\alpha \rightarrow t$ **ratio** vs. $c \dots$ critical value(s) under H_0
according to df and the H_1 type $\rightarrow$ reject/not reject H_0 .
- ▶ Through the **p -value**:
(significance level $\alpha \rightarrow$) t **ratio** $\rightarrow p$ -value for the given H_0 according
to df and the H_1 type ($\rightarrow$ reject/not reject H_0).
- ▶ Keep in mind the difference between '**not rejecting**' the null hypothesis
and 'accepting' it (incorrect)!
- ▶ For $n - k - 1 > 120$, it is often assumed $t_{n-k-1} \approx N(0, 1)$, which gives us
the '2(-sigma) rule' of thumb for $\alpha = 5\%$ for the **two-tailed test**.

t test using the t statistic

- ▶ **t statistic** is a generalized version of the t ratio following (1) under a generalized $H_0 : \beta_j = a_j$

$$t_{\hat{\beta}_j} \equiv \frac{\hat{\beta}_j - a_j}{se(\hat{\beta}_j)}.$$

- ▶ **t statistic** identifies independent variables statistically significantly different from a_j .
- ▶ For example, $H_0 : \beta_j = 1$ is very useful for constant elasticity models.
- ▶ Steps of the hypotheses testing are the same as for the t ratio above.

Hypotheses testing: Example

Independent Variables	Dependent Variable: <i>log(salary)</i>		
	(1)	(2)	(3)
<i>log(sales)</i>	.224 (.027)	.158 (.040)	.188 (.040)
<i>log(mktval)</i>	—	.112 (.050)	.100 (.049)
<i>profmarg</i>	—	-.0023 (.0022)	-.0022 (.0021)
<i>ceoten</i>	—	—	.0171 (.0055)
<i>comten</i>	—	—	-.0092 (.0033)
<i>intercept</i>	4.94 (0.20)	4.62 (0.25)	4.57 (0.25)
Observations	177	177	177
R-squared	.281	.304	.353

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Economic vs. statistical significance

- ▶ Check the statistical significance first: if significant, assess its economic/practical importance.
- ▶ Even if not statistically significant, we might still care about the expected effect of x on y , especially if it seems practically large: check the actual p -value whether it is not close to being significant at the given α .
- ▶ Also, think carefully about a potential bias!
- ▶ It is much more troublesome to have a significant variable with an unexpected sign and a practically large effect than an insignificant variable we thought would play a role.

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Confidence interval

- ▶ Under the CLM assumptions MLR.1 through MLR.6, we can easily construct a **confidence interval (CI)** for the population parameter β_j .
- ▶ Point vs. interval estimates.
- ▶ Using the distribution of $\hat{\beta}_j$: $\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$, we compute a $1 - \alpha$ **confidence interval** as

$$\boxed{\hat{\beta}_j \mp t_{n-k-1, 1-\alpha/2} se(\hat{\beta}_j)}.$$

- ▶ For $n - k - 1 > 120$, the '2(-sigma) rule' of thumb for $\alpha = 5\%$ can be again used for a rough idea.

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A specific type of the null hypothesis (readings: 4.4)

- ▶ We might be interested in testing $H_0 : \beta_1 = \beta_2$ rather than the commonly used $H_0 : \beta_1 = 0$.
- ▶ We need to transform/rewrite the null hypothesis into a 'testable' version, i.e., a linear combination $H_0 : \beta_1 - \beta_2 = 0$.
- ▶ This allows us to write the t statistic as

$$\boxed{t} \equiv \frac{\hat{\beta}_1 - \hat{\beta}_2 - 0}{se(\hat{\beta}_1 - \hat{\beta}_2)} = \boxed{\frac{\hat{\beta}_1 - \hat{\beta}_2}{se(\hat{\beta}_1 - \hat{\beta}_2)}}.$$

- ▶ What is the catch here?

Rewriting the model

- ▶ This issue can be overcome by using the rewritten null hypothesis $H_0 : \theta \equiv \beta_1 - \beta_2 = 0$ and substituting into the original model using $\beta_1 = \theta + \beta_2$.
- ▶ Example:

$$\begin{aligned}y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u = \\&= \beta_0 + (\theta + \beta_2)x_1 + \beta_2 x_2 + u = \\&= \beta_0 + \theta x_1 + \beta_2(x_1 + x_2) + u.\end{aligned}$$

- ▶ We can now estimate the model and test $H_0 : \theta = 0$.
- ▶ And what is the catch here now?

Seminars and the next lecture

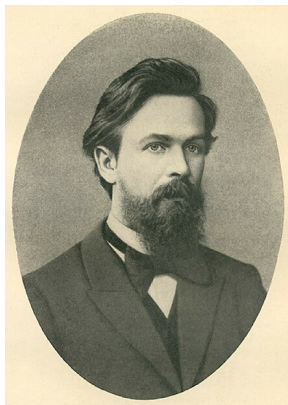
- ▶ HA #1 reminder:
 - ▶ deadline: Monday, March 23, 23:59:59,
 - ▶ delivered via the 'Study group roster' (Lecture JEB109) in SIS.
- ▶ Seminars:
 - ▶ hypotheses testing: single parameter (t test, p -value),
 - ▶ confidence intervals,
 - ▶ hypotheses testing: single linear combination,
 - ▶ discussing statistical vs. economic significance,
 - ▶ two computer exercises: hypotheses testing in R.
- ▶ Next lecture #6:
 - ▶ testing multiple linear restrictions: F test,
 - ▶ OLS asymptotics
 - ▶ consistency,
 - ▶ asymptotic normality and efficiency,
 - ▶ large sample inference.
- ▶ Readings for lecture #6:
 - ▶ Chapter 4: 4.5, 4.6, Chapter 5.

Historical spotlight: Andrey Markov (1856–1922)

Andrey A. Markov put dependence at the center of probability. Working in St. Petersburg, he pushed the Law of large numbers and the Central limit theorem beyond independent draws to cases where each step depends on the last. From this came what we now call a Markov chain: the next state depends on the current state, not the whole past. To make the idea concrete, he even turned to literature and tested the idea on Pushkin's Eugene Onegin. In 1913, he coded a long passage as vowels and consonants and counted how often one followed the other. The pattern showed clear dependence, exactly what his chains capture.

He also stands behind the 'Markov' in the Gauss-Markov theorem. Gauss had justified least squares much earlier, but in 1900, Markov gave a modern proof that does not require normality: under zero-mean, uncorrelated, constant-variance errors, Ordinary Least Squares is the 'Best linear unbiased estimator' (BLUE).

Why it matters for econometrics: many data sets have memory and do not reset each period. Markov's Chains let us model such serial dependence with transition probabilities, while Gauss-Markov theorem tells us when plain OLS is optimal without assuming normality. Together they give us practical tools for regime switches, credit transitions, customer turnover, and other problems where the present shapes the next step.



Source: [Wikipedia](#)

Historical spotlight: William Sealy Gosset (1876–1937)

William Sealy Gosset was a trained statistician and chemist who became the Head Experimental Brewer at Guinness in Dublin. His job was practical: improve beer quality and make it reliable using small test batches of barley and hops. Large-sample methods were not applicable in his experiments, so he went to University College London to study with Karl Pearson, then brought the ideas back to the brewery and tested them on real data from beer production.

From this work came the 'Student' t test and the t distribution. Gosset showed how to compare means and set uncertainty when samples are small and variance is unknown. His tools let Guinness choose better barley with far fewer cross-breeding trials and they now sit at the heart of econometrics.

And why is it called 'Student' and not 'Gosset'? Guinness did not allow staff to publish under their own names to protect company know-how. Gosset persuaded them to let him publish his general statistical results only if he used a pseudonym. So he wrote as 'Student' and published most of his 21 academic papers under that name. The label stuck, which is why we still say Student's t today.

Takeaway for econometrics: design small, careful experiments and use small-sample tools when data are scarce. The t test and t confidence intervals provide honest uncertainty without assuming large samples, which is exactly what many real problems need.



Source: [Wikipedia](#)